

Weather-Driven Power Outages

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Major Weather Drivers of Outages:

High wind/downed lines: During rain storms, the water retention of different soil types can affect the stability of the utility poles; when the soil is saturated and soft, poles are more likely to lean or topple in high winds, resulting in grid instability (Lee et al., 2025)

Ice/freezing rain accretion on lines and towers: The accretion of precipitation, in-cloud droplets, or hoarfrost on the surface object. Recently, widespread power outage events have resulted from extreme atmospheric icing on electricity transmission and distribution networks caused by ice storms (Cerrai et al., 2020).

Lightning: Causes power outages by overwhelming the electrical insulation on power lines, creating short circuits that force protective equipment to shut down the circuit. A single bolt carries tens of thousands of amps and can spike voltages far beyond what power line components are designed to handle. Even strikes that miss power infrastructure entirely can induce dangerous surges on nearby lines from up to 50 meters away (scienceinsight.org, 2026).

Extreme heat (thermal derating, demand spikes): Mostly caused by a "double-whammy" of situations; during extreme heat situations (i.e., heat waves) there's an increased demand and stress infrastructure. When temperatures rise, millions of people simultaneously used their air conditioners, creating immense pressure on the power grid. The rise of demand coincides with conditions that weaken the grid's ability to deliver. For example, during high temperatures the transmission lines sag, more likely to come in contact with trees or other objects, leading to short circuits and outages ([IERE Team](#), 2025).

Wildfire (PSP events): During wildfires there are two likely situations that cause power outages. First, from extreme weather events like high winds can cause damage and cause wildfires to grow; the second, is caused by Public Safety Power Shutoffs (PSPS), where utility companies shut off power to prevent electrical equipment from starting fires – especially during high winds and dry conditions. PSPS events are necessary to prevent any additional fire activity (Mahapatra, 2025).

Flooding: Water into any electrical systems, both above and below ground, can create short circuits, damage equipment, and force companies to shut down power to entire areas to prevent electrocution and further damage. Floodwater is very conductive from the dissolved minerals within it such as, salts, chemicals, and debris which could be a dangerous pathway for electricity ([IERE Team](#), 2025).

Basic Grid Vulnerability:

Transmission exposure: Angular winds on the transmission power lines can lead to overloading the lines and exposing the transmission tower the risk of causing a power outage (Greco et al., 2026).

Distribution exposure: The infrastructure of distribution lines are mostly installed along streets and are exposed to the elements of intense storms. Primarily strong winds that can knock down the poles (Blumsack et al., 2026).

Vegetation/tree-related failures: During major storms, distribution lines and poles fail mainly due to fallen trees rather than the direct impact of wind (energy.gov, 2026).

Cascading failures: Is a form of failure in a system comprising interconnected parts, as in the failure of a part can trigger the failure of successive parts. This form of cascading failures can happen in computer networks and power grid systems (Mahmoud et al., 2019). There are various causes for how a cascading failure can happen in a power system, the main four kind are natural disasters, human activity, component failure, and system failures (Guo et al., 2017).

Outage Prediction Modeling Approaches:

Statistical/regression models:

Simple Linear Regression: A statistical method to how a single independent variable influences a dependent variable. Identifies the straight line that best represents the relationship between the two variables within the data. This line can then be used to predict the values of one variable given the values of the other (Lee et al., 2025).

Multiple Linear Regression: is a statistical model that analyzes the simultaneous effects of various independent variables on a dependent variable. Models the relationship between several predictors and the outcome, enabling the construction of a predictive model in which each independent variable may have a distinct influence on the dependent variable (Lee et al., 2025).

Ridge Regression: A statistical model that employs a technique to address issues arising from a high degree of *multicollinearity* among independent variables or excessive model complexity leading to overfitting.

Multicollinearity: The phenomenon in which two or more independent variables in a regression analysis exhibit a strong linear correlation with each other.

Regularization is a term that penalizes significant regression coefficients, Ridge Regression constrains the magnitude of these coefficients, thereby encouraging a more parsimonious (unwilling to spend more resources than needed) model structure. The benefit of regularization facilitates more stable and reliable parameter estimation and, consequently, enhances the predictive performance of the model (Lee et al., 2025).

Lasso Regression: A statistical model that uses a regularization technique designed to prevent overfitting while simultaneously enabling automatic variable selection by shrinking the coefficients of less significant or irrelevant predictors to precisely zero. Like Ridge Regression, Lasso Regression imposes a penalty on the magnitude of the regression coefficients; however its unique ability to reduce some coefficients to zero facilitates the identification and retention of only the most significant variables. Resulting in a sparser and more interpretable model, enhancing both model simplicity and interpretability (Lee et al., 2025).

Machine-learning outage prediction (weather features --> outage counts):

K-Fold Cross-Validation: enables a robust evaluation of the model's ability to generalize. With higher k-values in a k-fold cross-validation, it does not significantly improve model performance. So, for large scale multivariable observations it's best to have the k-value as 2-10 folds (Lee et al., 2025).

Bayesian Optimization: Used to optimize the hyperparameters of multiple machine learning models in effort towards selecting the model. It's a common technique used to maximize the performance of machine learning models (Lee et al., 2025).

Partial Dependence Plot (PDP): A graph that visualizes the average effect of a specific input feature on the predicted outcome of a machine learning model. PDP is particularly useful for interpreting models that are difficult to understand or are considered "black-box", such as

ensemble models, by helping to reveal the internal decision-making process (Lee et al,2025).

Random Forest (RF): Formed by 200 decision trees. Each decision tree is able to split the dataset into subsets for which the response variable is similar, by minimizing the sum of the square error. Each tree of the RF is trained on a random subset of training data using a random subset of explanatory variable (Cerrai et al., 2020).

Bayesian Additive Regression Tree (BART): Created through statistical sum of 30 Bayesian trees, that are the base learners. A Markov Chain Monte Carlo (MCMC) algorithm is used with 10,000 iterations for fitting the BART model, and 4,000 iterations are used for obtaining the predictions (Cerrai et al., 2020).

Reliability metrics:

SAIDI: System Average Interruption Duration Index; for a power system, measures the average outage duration (typically over a year) for each customer. Can be mathematically expressed as the following (Becker, 2024):

$$\text{SAIDI} = (\text{Sum of all customer interruption durations}) / (\text{Total number of customers served})$$

Lower values of SAIDI = higher reliability and better service

Higher values of SAIDI = intense reliability issues on the service and/or the power system

SAIFI: System Average Frequency Index; for a power system, measures the average number of interruptions a customer experiences over a specific period (typically a year). Can be mathematically expressed as the following (Becker, 2024):

$$\text{SAIFI} = (\text{Total number of customer interruptions}) / (\text{Total number of customers served})$$

SAIFI = 0 ; ideal meaning that interruptions are rare

Higher values of SAIFI = insufficient maintenance and service from the power system

CAIDI: Customer Average Interruption Duration Index; the average time to restore service. Can be mathematically expressed as the following (Michigan Public Service Commission, 2026)

$$\text{CAIDI} = (\text{total minutes of customer interruption}) / (\text{total number of interruptions})$$

Lower values of CAIDI = Faster the utility is restored to service customers.

Higher values of CAIDI = Slower maintenance time to restore service to customers.